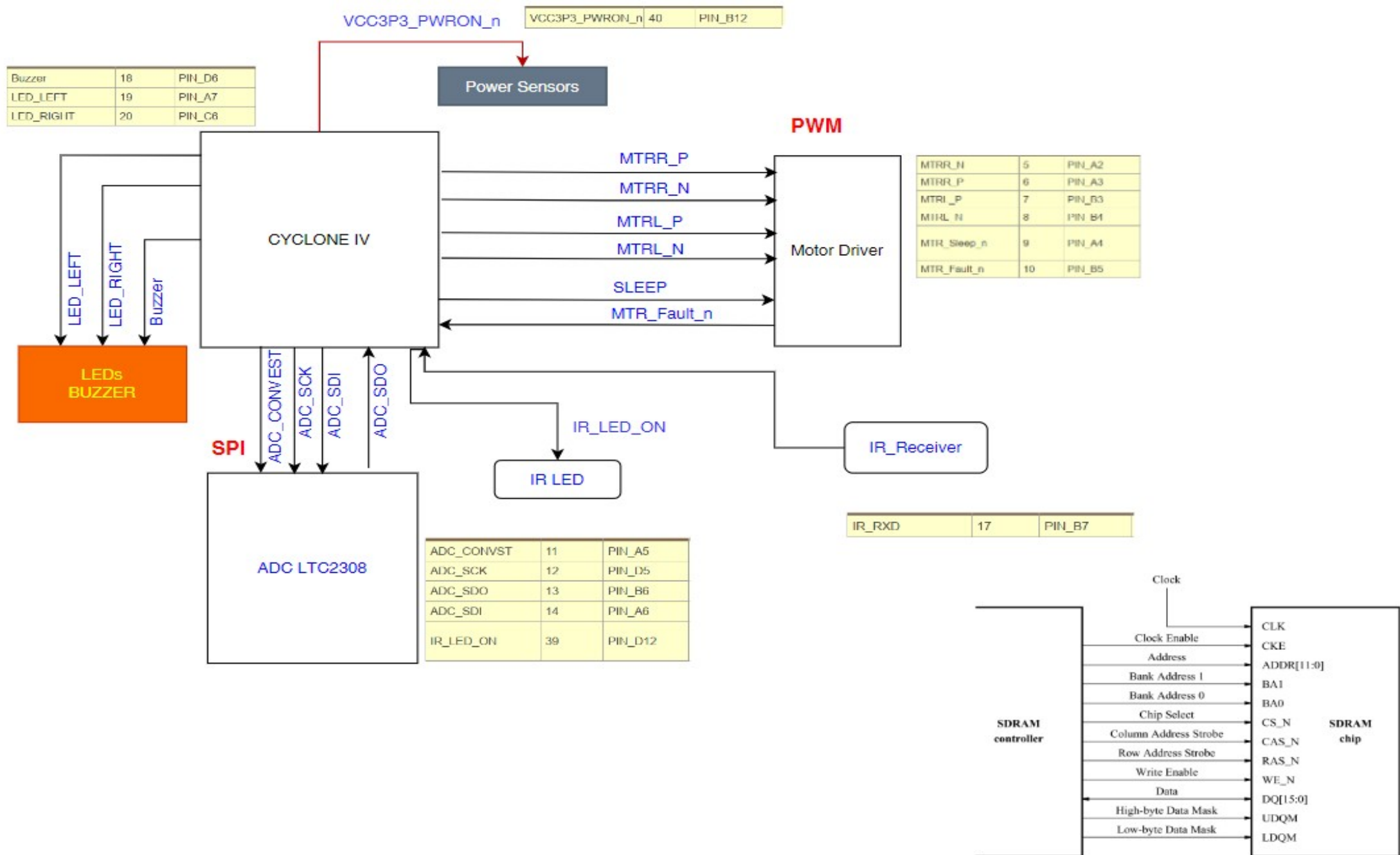
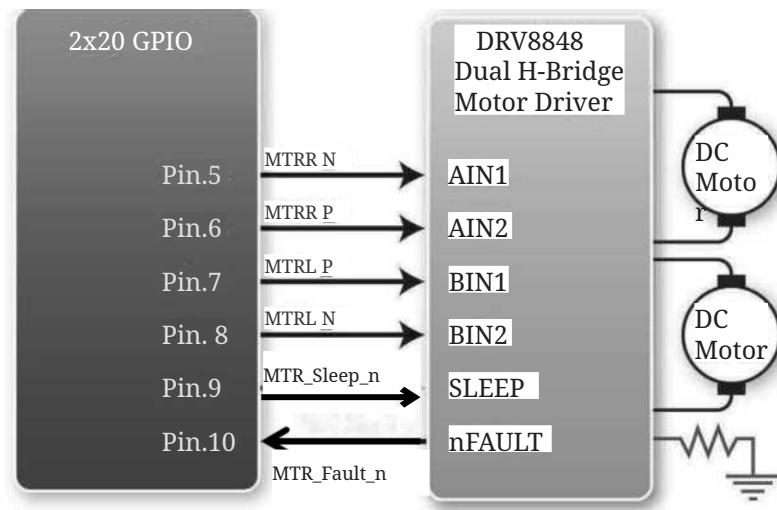


TP A-Cute Car

Guide



Pilotage Moteur

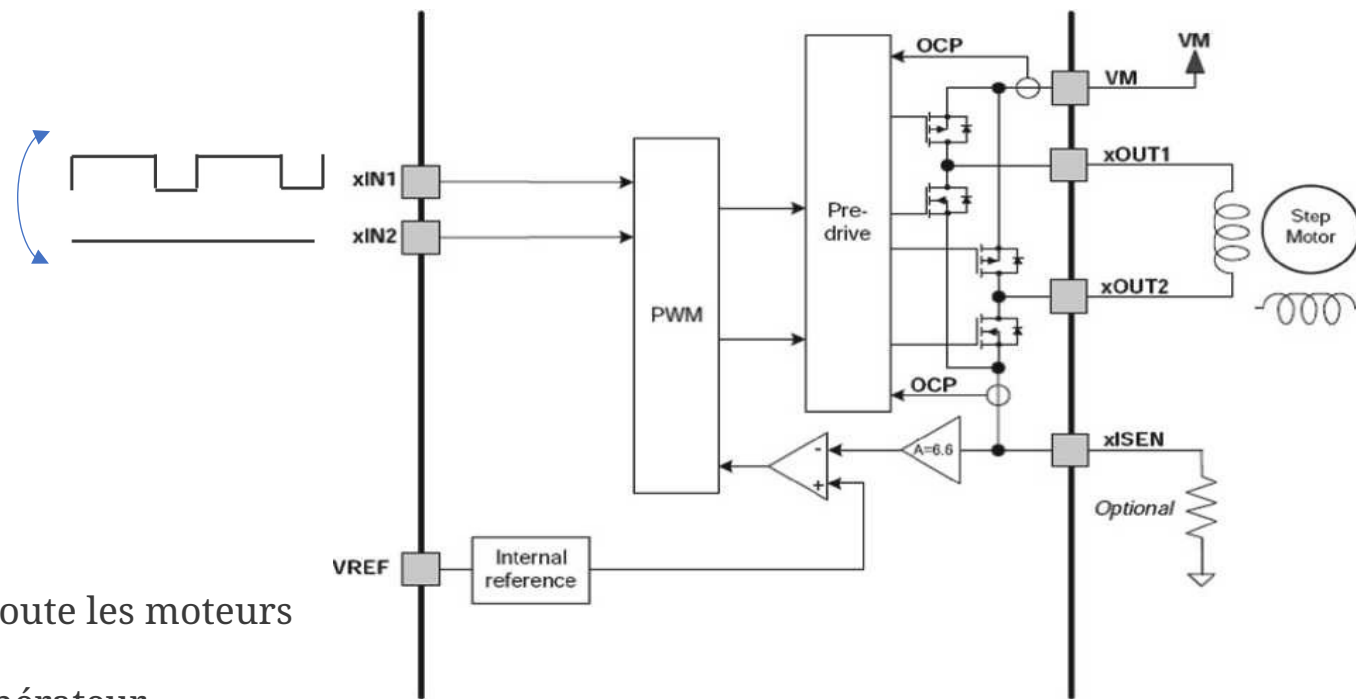


Signal **MTR_Sleep_n** à positionner à 1 pour mettre en route les moteurs

Signaux **MTRx_N** et **MTRx_P** à générer à l'aide d'un générateur PWM

MTRR_P	MTRR_N	MTRR_OUT_Ap	MTRR_OUT_An	Function (DC Motor)
MTRL_P	MTRL_N	MTRL_OUT_Bp	MTRL_OUT_Bn	
0	0	Z	Z	Coast (fast decay)
0	1	L	H	Reverse
1	0	H	L	Forward
1	1	L	L	Brake (slow decay)

Circuit de commande des moteurs Ponts en H - DRV8848



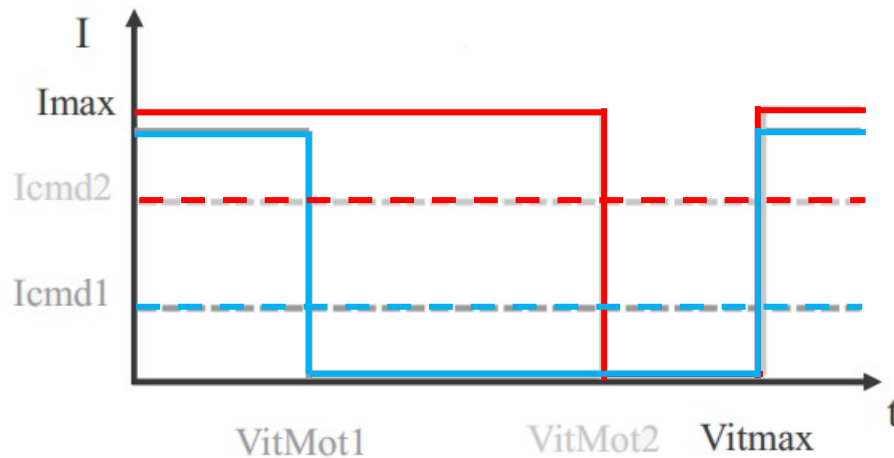
Driver moteur génère les signaux de commande du pont en H à partir des signaux PWM en entrée

Contrôle vitesse moteurs par Modulation à Largeur d'impulsion

Commande d'un moteur

Go	Dir	VitMot(12bits)
----	-----	----------------

Go = 1 : Marche Dir = 1 : Arrière
Go = 0 : Arrêt Dir = 0 : Avant



$$I_{cmd} = I_{max} * (VitMot/Vitmax)$$

La vitesse du moteur est proportionnelle à Icmd

Calcul de la vitesse

$$VitMotD = VitMoy + pos_ligne * K$$

$$VitMotG = VitMoy - pos_ligne * K$$

Si DirMotG = DirMotD = 0

Si VitMotD = VitMotG

Robot va tout droit

Si VitMotD > VitMotG

Robot tourne vers la gauche

Si VitMotD < VitMotG

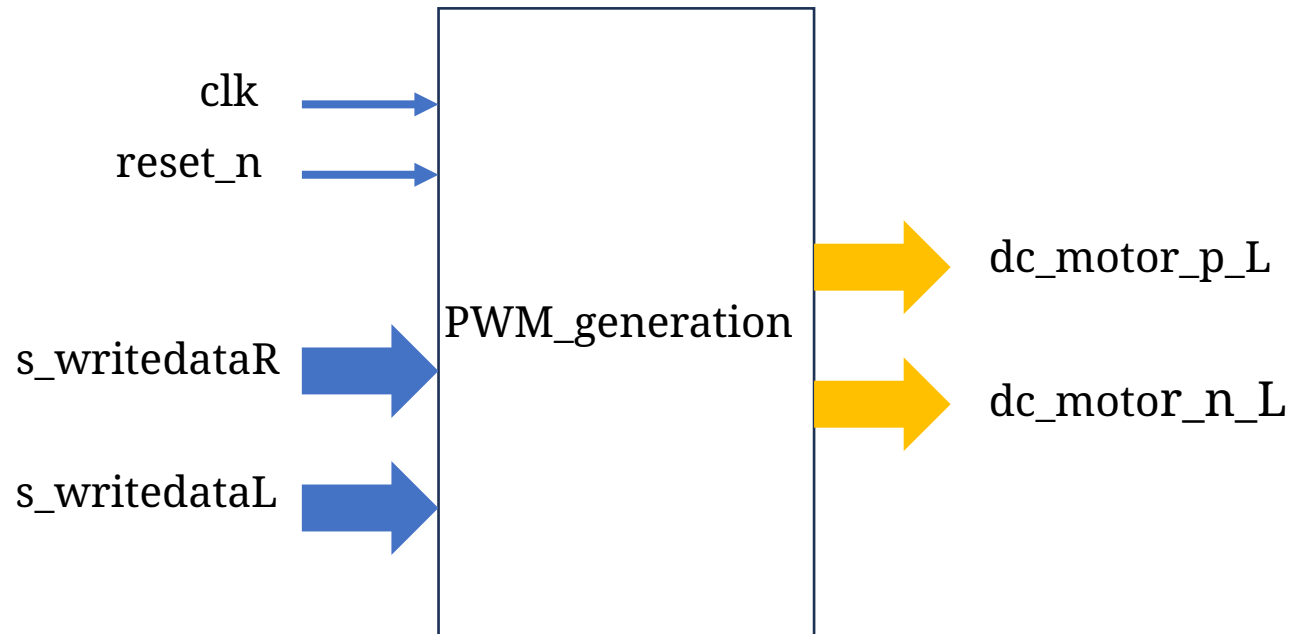
Robot tourne vers la droite

Si DirMotG = 0 et DirMotD = 1

Si VitMotD = VitMotG Robot fait

Robot fait demi-tour sur place

Module VHDL générateur PWM

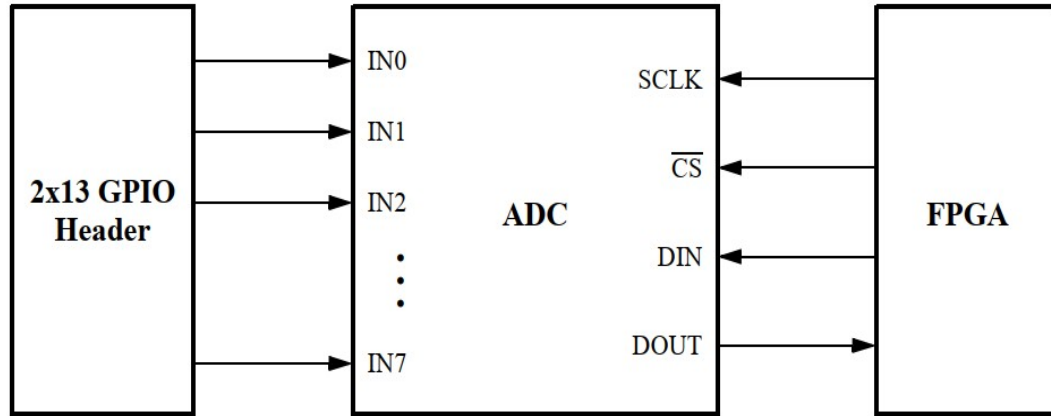


```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;
library lpm;
use lpm.all;

-----
-- FREQUENCE FPGA de la DEO-NANO: 50 MHZ ----
-----

entity PWM_generation is
    port(
        clk,reset_n:in std_logic;
        -- Le bit13 : bit de go(1)/stop(0).
        -- Le bit12: bit de forward(0)/backward(1).
        -- Les bits 11 à 0: vitesse=durée état haut
        s_writedataR,s_writedataL: in std_logic_vector(13 downto 0);
        dc_motor_p_R,dc_motor_n_R, dc_motor_p_L,dc_motor_n_L: out std_logic
    );
end entity;
```

ADC de la DE0-NANO : ADC128S002



Signals to and from the ADC

8 analog input pins IN0 through IN7. ADC reads and convert the signal on one of these eight input channels

SCLK serves as a device clock

CS select the ADC chip.

The DIN and DOUT wires are used for transferring addresses and data between the two chips.

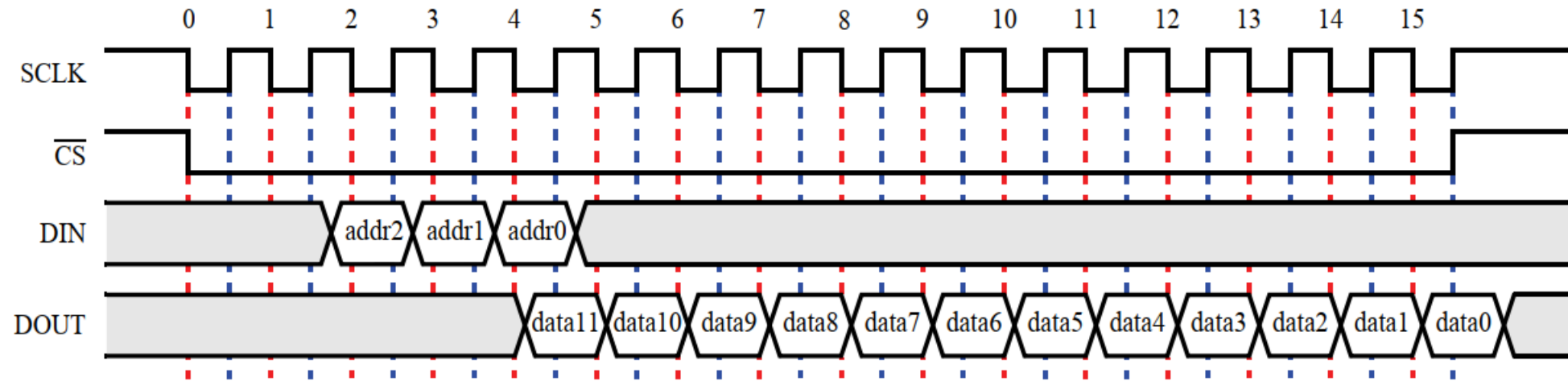
DIN is mapped to the ADC_SADDR. Provide the address of the **next channel requested for conversion.**

The address is 3 bits in length and is sent to the ADC serially at a rate of 1 bit per SCLK cycle.

DOUT is mapped to the ADC_SDAT and used to send the digital value (12 bits long) of the converted signal.

Sent to the FPGA in a serial manner at a rate of 1 bit per SCLK cycle

Timing



Timing requirements for the ADC

A réaliser

A- On se propose d'étudier les deux modules :

- l'IP Core fourni par Altera interfaçant l'ADC ADC128S002 au bus Avalon (cartes DE0-nano) (module 1)
 - le bloc VHDL capteurs_sol_seuil (module 2) pilotant l'ADC LTC2803
- 1- Implémenter et mettre en oeuvre le module 1 (Acquérir deux signaux 0 et 3.3 v)
 - 2- Transformer le module 2 en IP core pour qu'il soit interfacé au bus Avalon (Acquérir deux signaux 0 et 3.3 v).
 - 3- En utilisant l'analyse logique, retrouver les chronogrammes de fonctionnement de :
 - l'ADC LTC 2803
 - l'ADC ADC128S002

B – Comment peut-on caractériser la valeur du seuil à fixer pour le module 2

C- Transformer tous les modules fournis sous forme d'IP Core. Déduire un système exclusivement fait d'IP Core pouvant réaliser un suivi de ligne.

ADC Controller

<i>DE0-Nano ADC Controller register map</i>			
Offset in bytes	Register name	Read/Write	Purpose
0	CH_0	R	Converted value of channel 0
	Update	W	Update the converted values
4	CH_1	R	Converted value of channel 1
	Auto-Update	W	Enables or disables auto-updating
8	CH_2	R	Converted value of channel 2
12	CH_3	R	Converted value of channel 3
16	CH_4	R	Converted value of channel 4
20	CH_5	R	Converted value of channel 5
24	CH_6	R	Converted value of channel 6
28	CH_7	R	Converted value of channel 7

The **Update register** is used to initiate a conversion operation. Performing a write to this register will update all channels in use, with the new values becoming available once the entire conversion process has finished. If reads to the channel registers are attempted while a conversion is taking place, then the wait_request signal will be raised, causing the processor to stall until the update has finished.

The **Auto-Update register** is initially loaded with a zero value. The auto-update allows the system to automatically begin a second update cycle after the first finishes. As result, channel values can be accessed during an update cycle, and it is user's responsibility to ensure the values used are up-to-date. Writing a '1' to this register enables auto-update, while writing a '0' disables it.

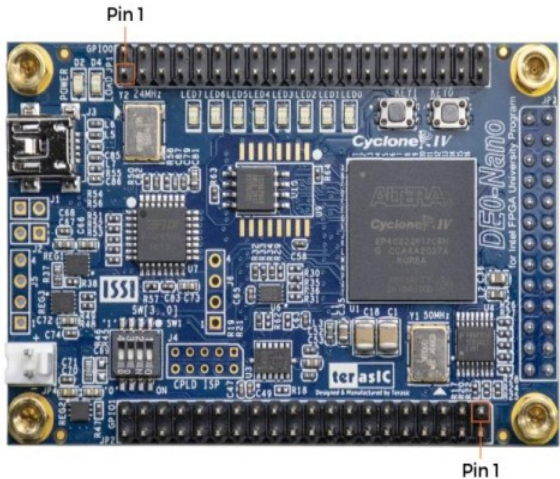
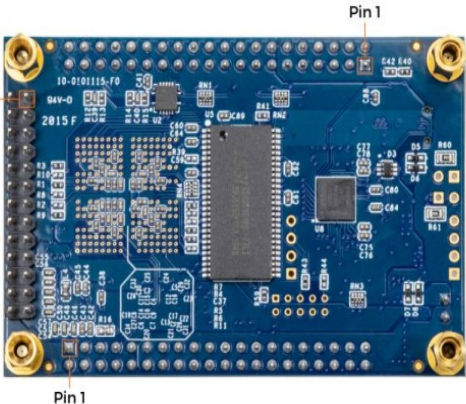
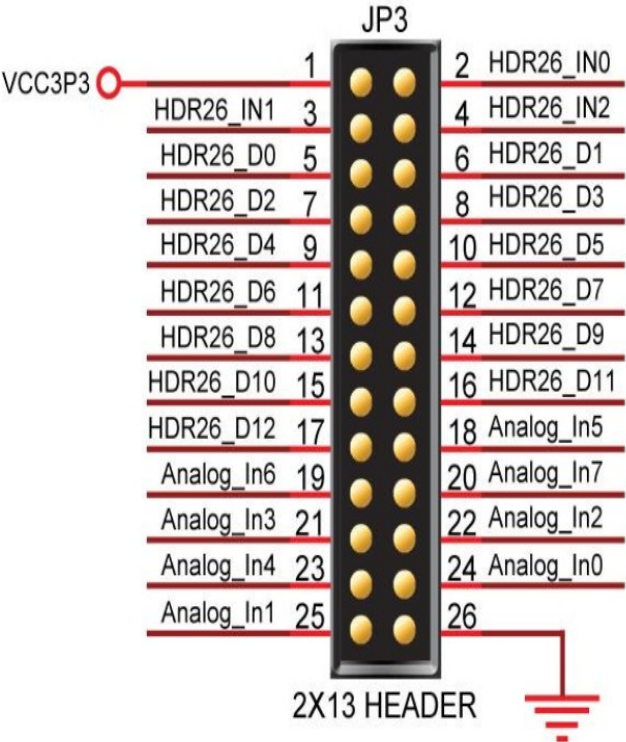
Registres ADC

Register Index	Register Name	Description	Read/Write
0	CS	Write: Bit 0 presents start bit, triggered by rising edge. Writing 0 then 1 to bit 0 start adc conversion. Read: Bit 0 presents read flag. Value 1 meansadc conversion is done and channel 0~7 data are ready on register 1~8.	RW
1	CH0	12 bit digitized value for channel 0	R
2	CH1	12 bit digitized value for channel 1	R
3	CH2	12 bit digitized value for channel 2	R
4	CH3	12 bit digitized value for channel 3	R
5	CH4	12 bit digitized value for channel 4	R
6	CH5	12 bit digitized value for channel 5	R
7	CH6	12 bit digitized value for channel 6	R
8	CH7	12 bit digitized value for channel 7	R

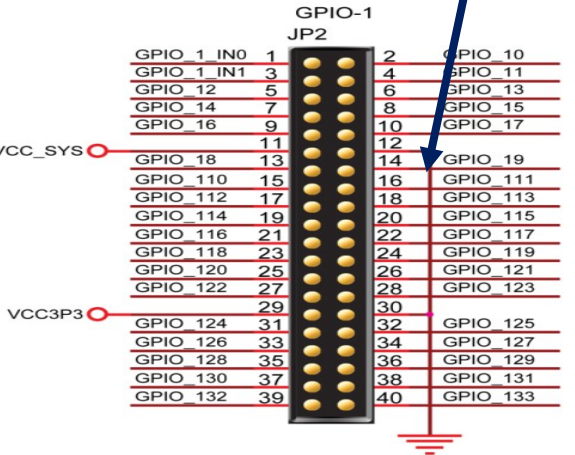
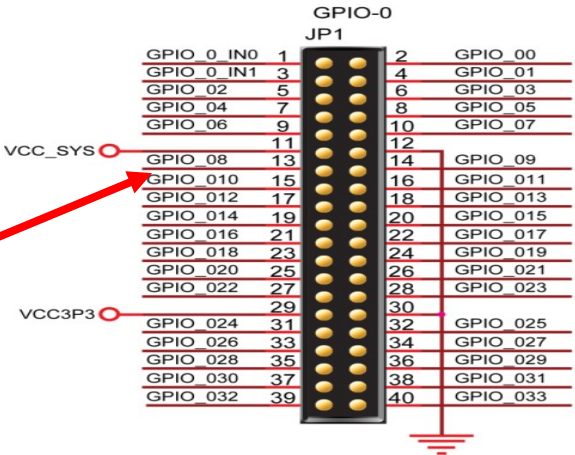
ADC DE0 NANO

Table 3-9 Pin Assignments for ADC

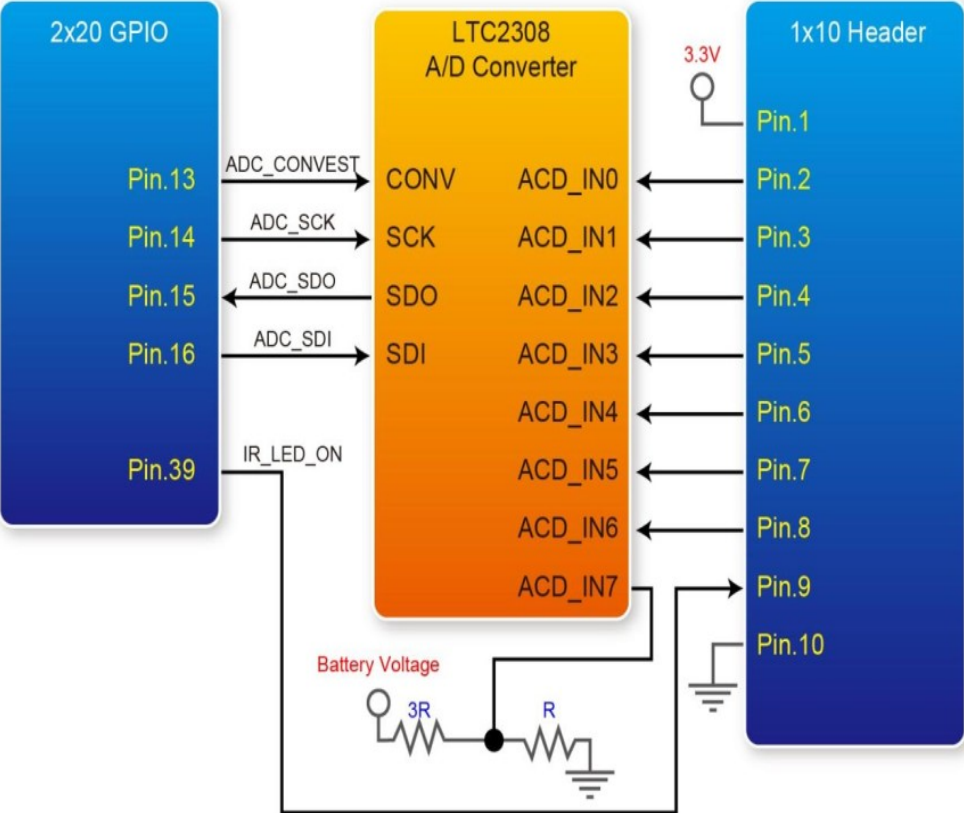
Signal Name	FPGA Pin No.	Description	I/O Standard
ADC_CS_N	PIN_A10	Chip select	3.3V
ADC_SADDR	PIN_B10	Digital data input	3.3V
ADC_SDAT	PIN_A9	Digital data output	3.3V
ADC_SCLK	PIN_B14	Digital clock input	3.3V



3,3v

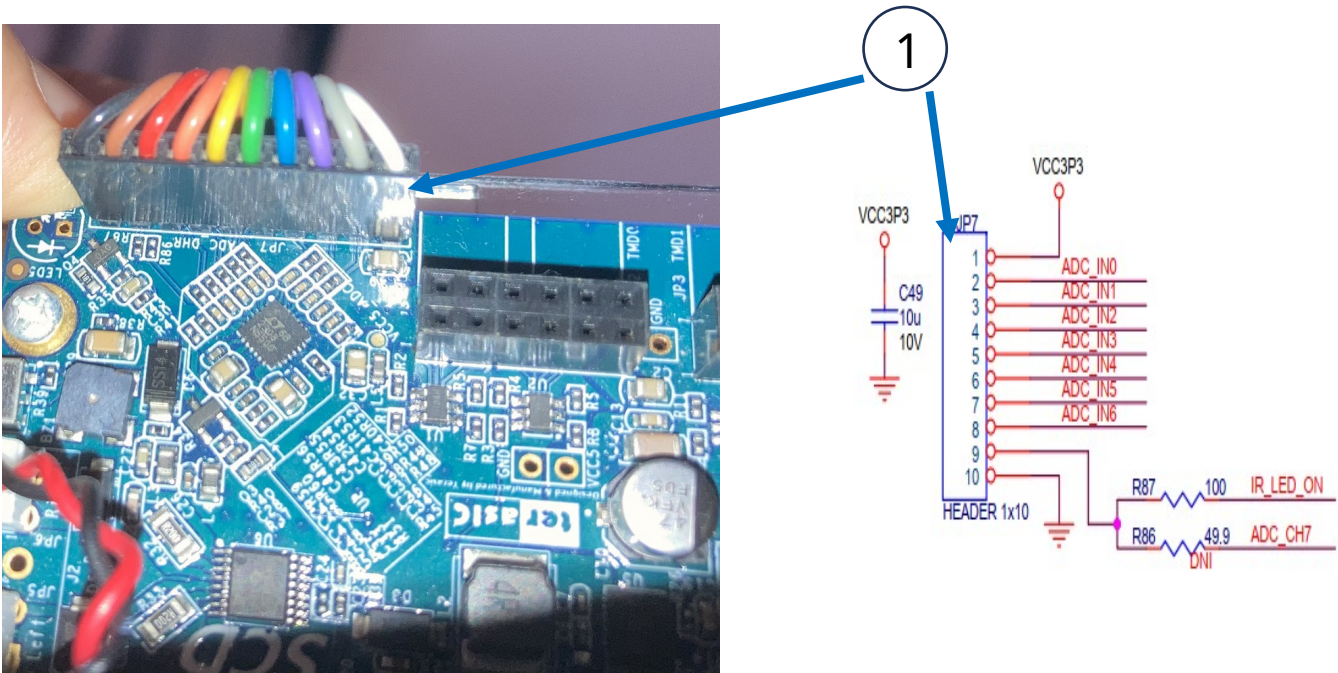


ADC A-Cute Car

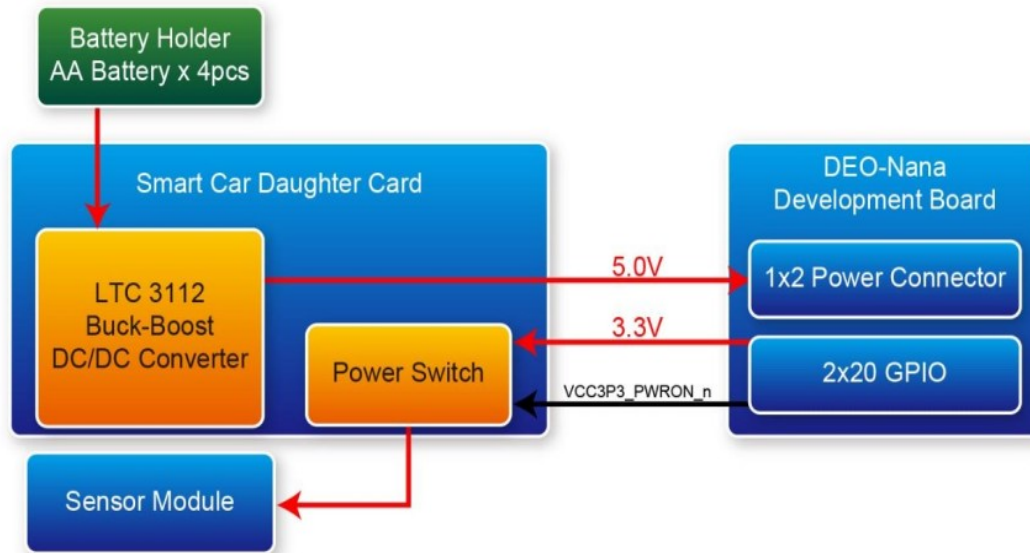


Pin assignment of ADC 4-Wire SPI interface and Power control

SCD Signal Name	GPIO Pin No.	FPGA Pin No. on GPIO 0	Description	I/O Standard
ADC_CONVST	11	PIN_A5	Right Motor Negative signal.	OUT, 3.3V
ADC_SCK	12	PIN_D5	Right Motor Positive signal.	OUT, 3.3V
ADC_SDO	13	PIN_B6	Left Motor Positive signal.	IN, 3.3V
ADC_SDI	14	PIN_A6	Left Motor Negative signal.	OUT, 3.3V
IR_LED_ON	39	PIN_D12	IR LED Power Control signal. Logic high to turn on IR LED power.	OUT, 3.3V



Alimentation capteur



Pin Assignment of 3.3V Power Control

SCD Signal Name	GPIO Pin No.	FPGA Pin No. on GPIO 0	Description	I/O Standard
VCC3P3_PWRON_n	40	PIN_B12	SCD 3.3V Power Control signal. Low Active.	OUT, 3.3V

LTC 2308

Configuration des canaux

S/D	0/S	S1	SO	0	1	2	3	4	5	6	7	COM
0	0	0	0	+	-							
0	0	0	1			+	-					
0	0	1	0					+	-			
0	0	1	1							+	-	
0	1	0	0	-	+							
0	1	0	1			-	+					
0	1	1	0					-	+			
0	1	1	1							-	+	
1	0	0	0	+								-
1	0	0	1			+						
1	0	1	0					+				-
1	0	1	1							+		-
1	1	0	0		+							-
1	1	0	1				+					-
1	1	1	0						+			-
1	1	1	1								+	-

Rôle des 6 bits SDI

S/D	0/S	S1	SO	UNI	SLP
-----	-----	----	----	-----	-----

S/D = SINGLE-ENDED/DIFFERENTIAL BIT

0/S = ODD/SIGN BIT

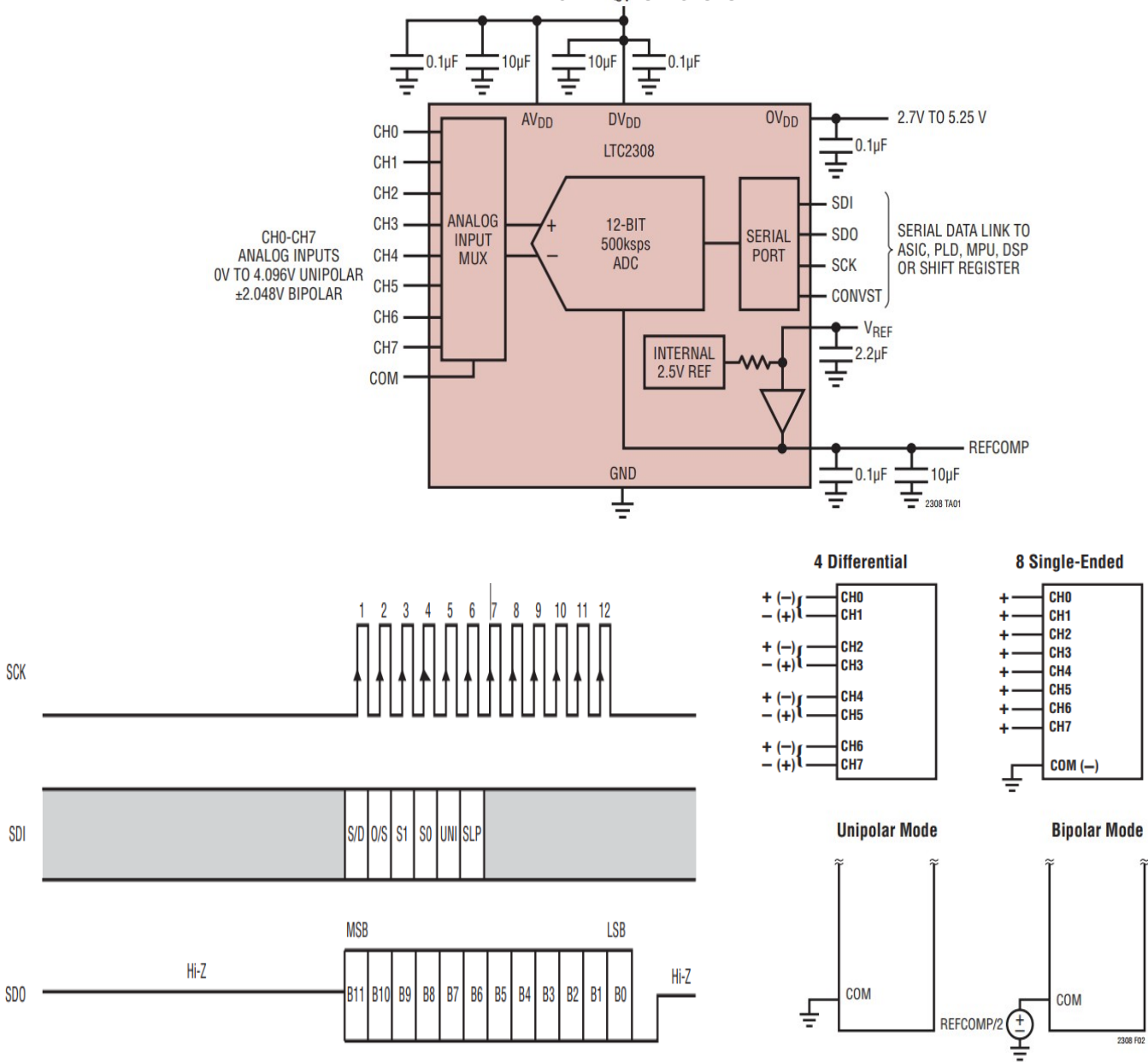
S1 = ADDRESS SELECT BIT 1

SO = ADDRESS SELECT BIT 0

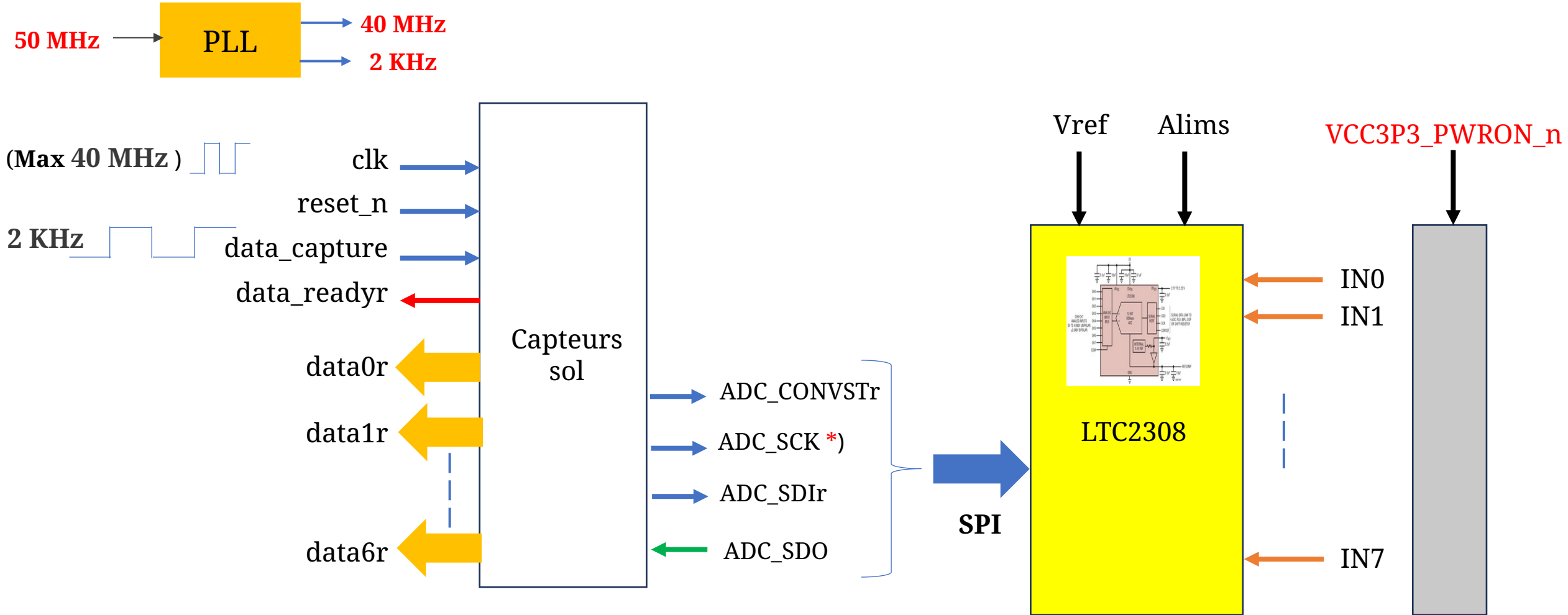
UNI = UNIPOLAR/BIPOLAR BIT

SLP = SLEEP MODE BIT

Possibilité de convertir des signaux référencés à la masse ou différentiels

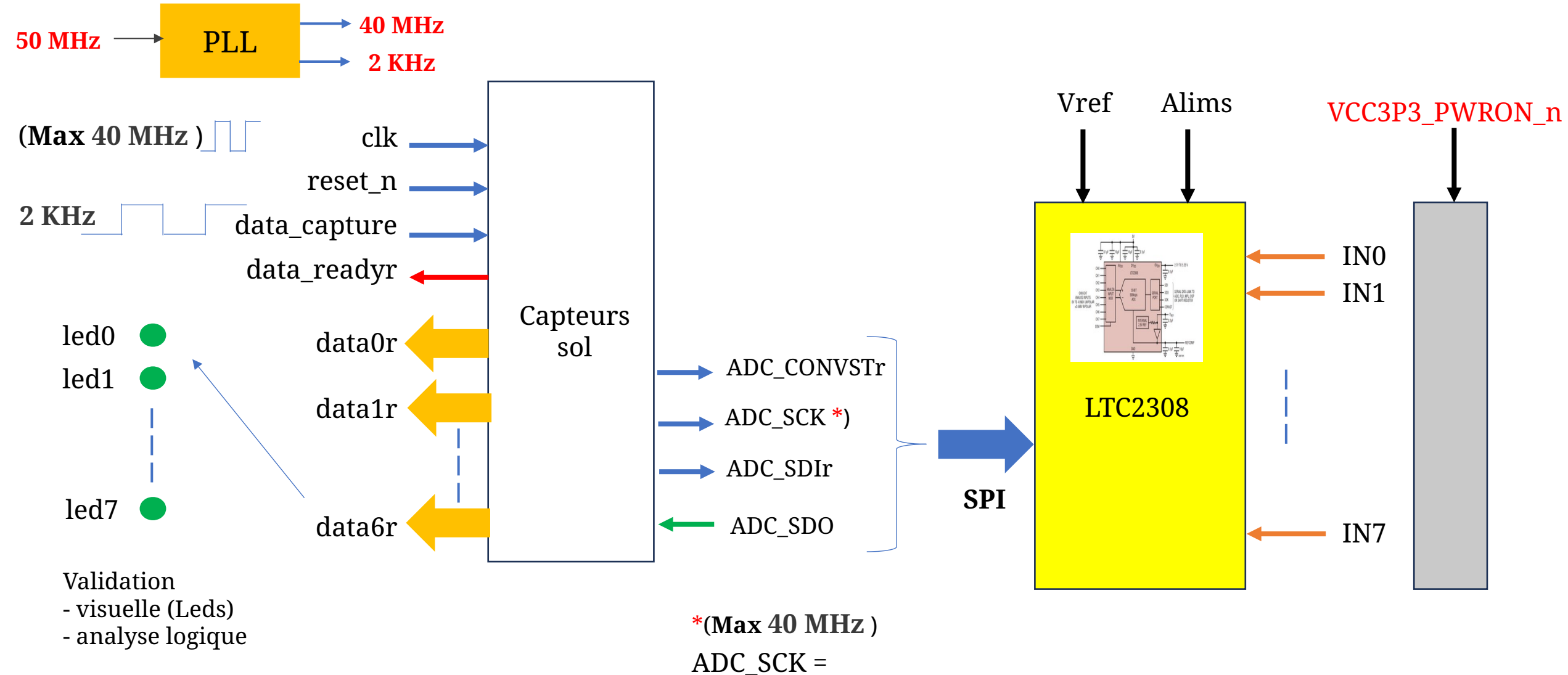


Module VHDL capteurs sol

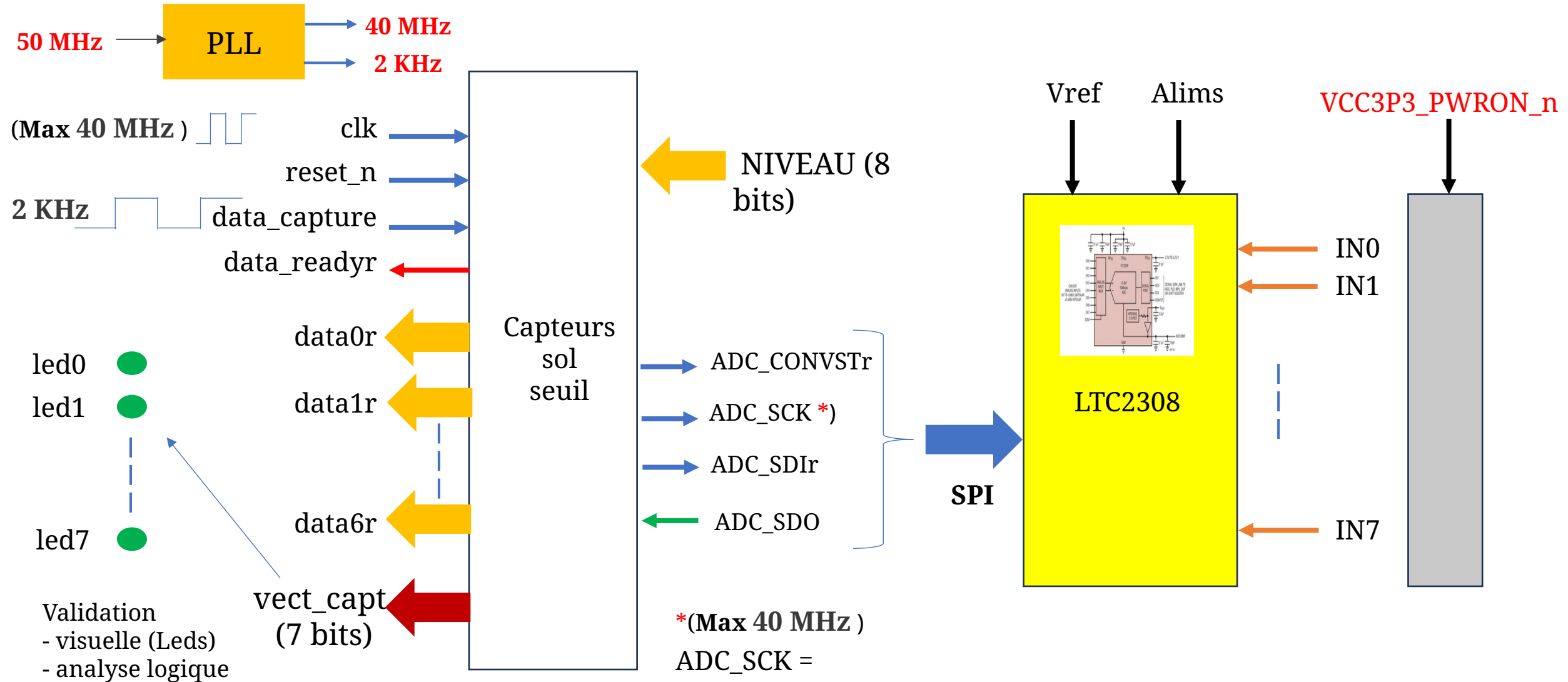


*(Max 40 MHz)
ADC_SCK =

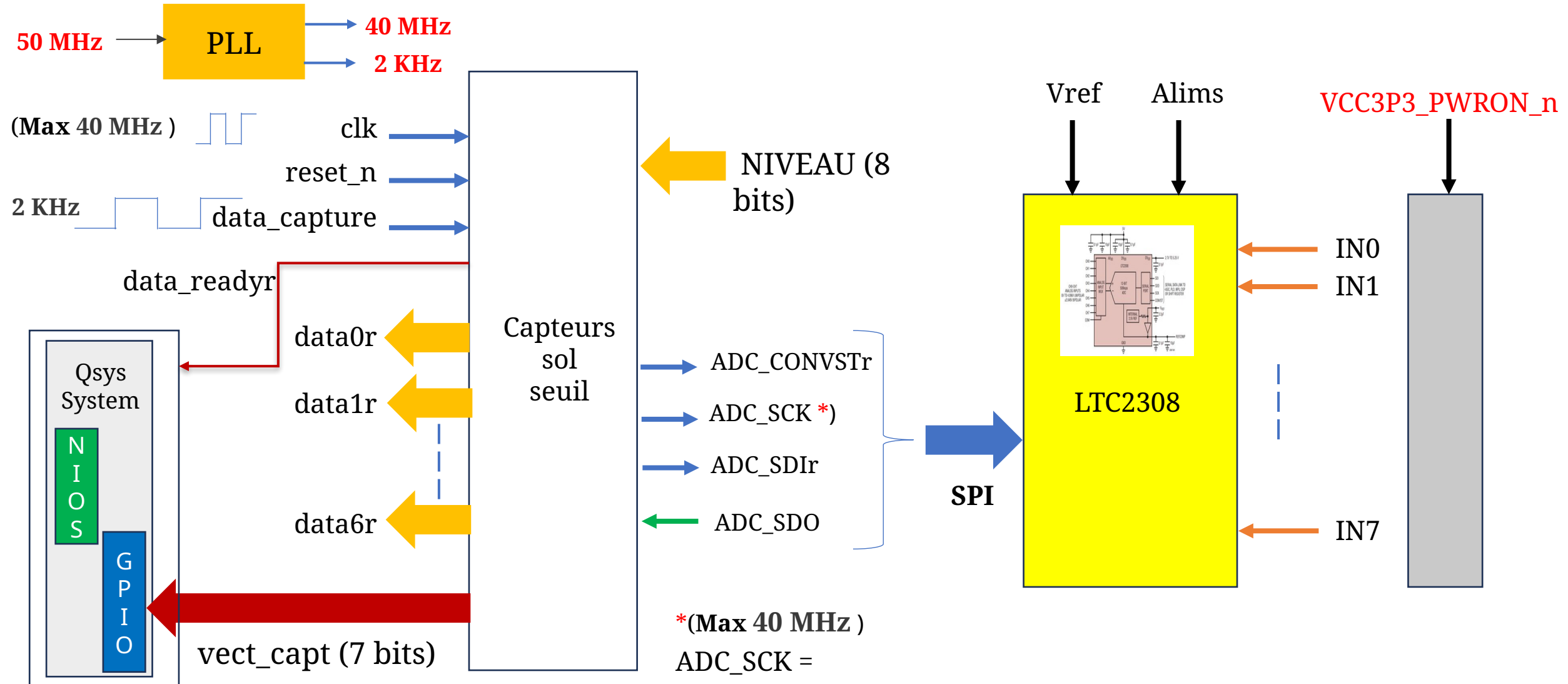
Module VHDL capteurs sol



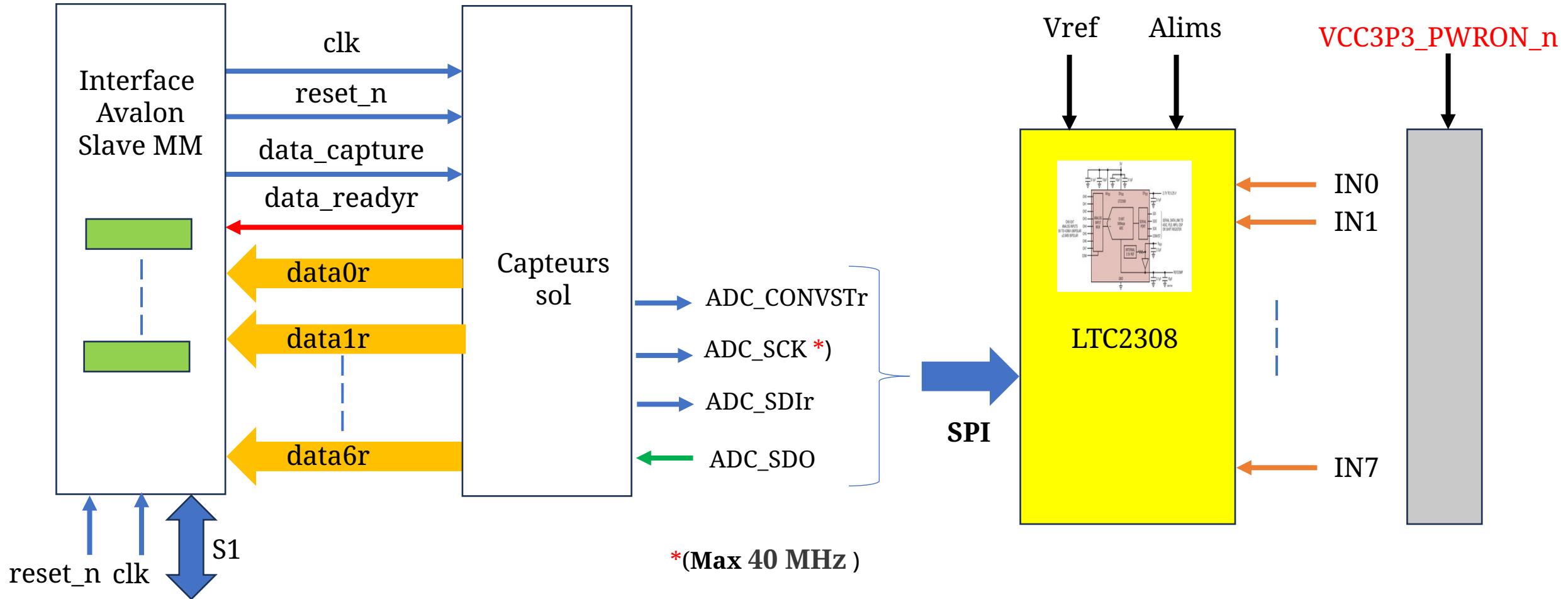
Module VHDL capteurs sol seuil



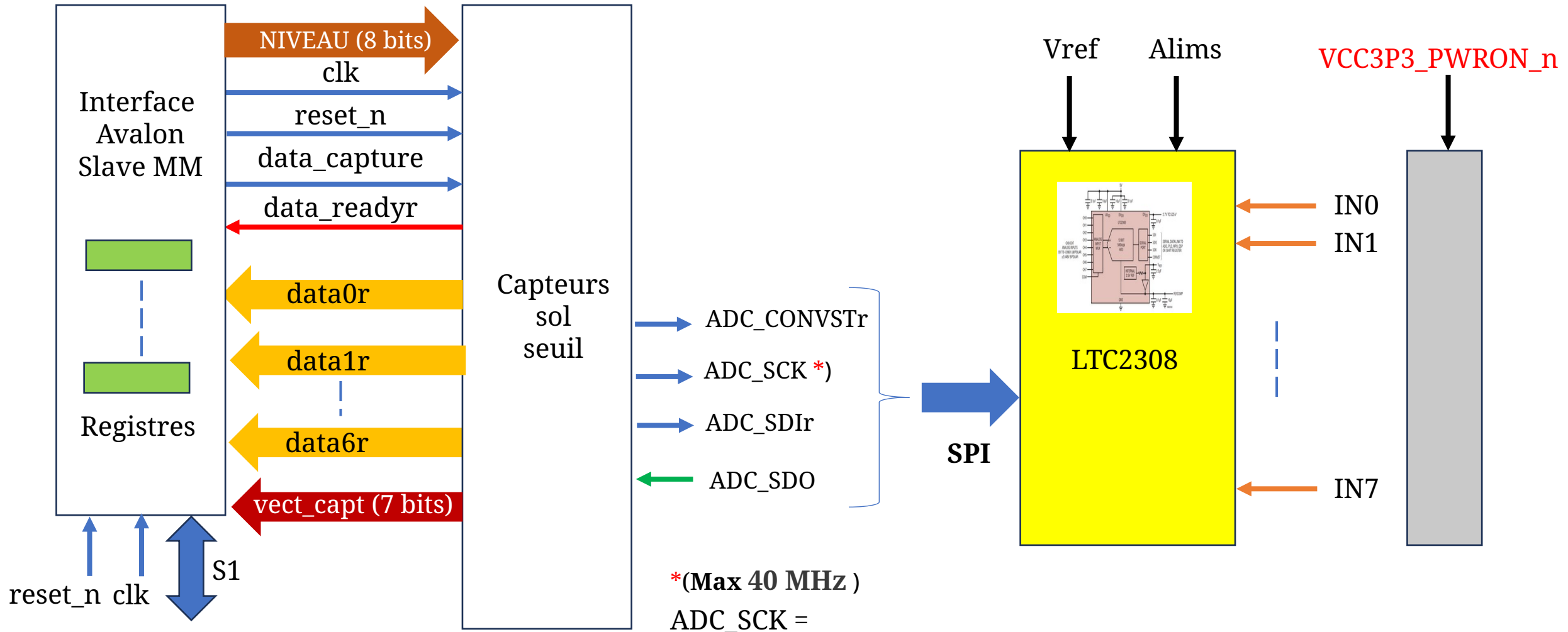
Module VHDL capteurs sol seuil



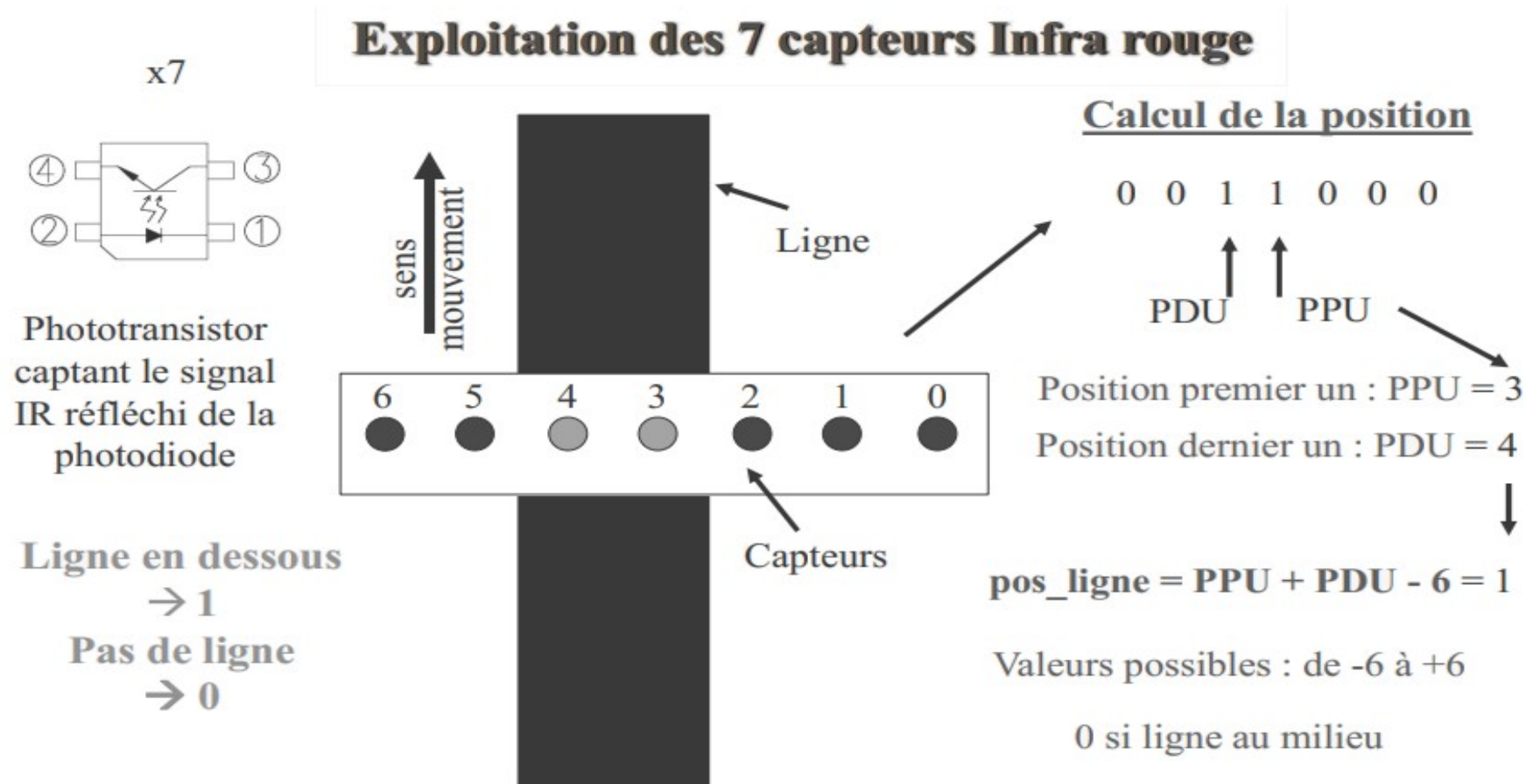
Module VHDL capteurs sol Avalon Slave MM



Module VHDL capteurs sol seuil Avalon MM



Position/ligne suivant valeurs capteurs



Machine à états

- Signaux d'état
 - Start_SL
 - Start_rot
 - Fin_SL
 - Fin_rot
 - Dir_rot

PID

```
integral = integral + error;  
derivative = error - last_error;  
last_error = error;  
output = (kp * error + ki * integral + kd * derivative); // PID  
turn = output * 100.0;  
LeftSpeed = Speed + turn;  
RightSpeed = Speed - turn;
```

Kp=1 et Kd = 0.8